Pythagoras' Theorem

Name _____ Date _____ Calculator allowed Version 1

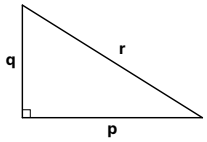
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KEY WORDS hypotenuse *(tapa roa)* · right-angled triangle *(tapatoru hāngai)* · square root *(pūtakerua)*

A The hypotenuse ●○○

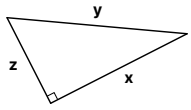
1. Write the letter of the hypotenuse of each triangle.

a)



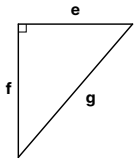
Hypotenuse: _____

b)



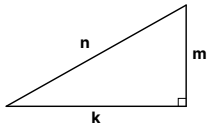
Hypotenuse: _____

c)



Hypotenuse: _____

d)



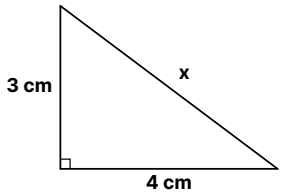
Hypotenuse: _____

B Finding the hypotenuse ●○○

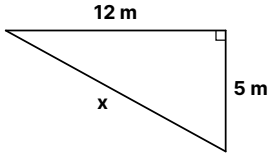
> **Remember:** $a^2 + b^2 = c^2$, where c is the hypotenuse, the longest side.

2. Find the hypotenuse, x. Round to 1 decimal place where needed. Diagrams are not to scale.

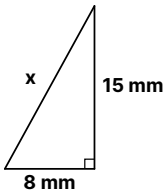
a)



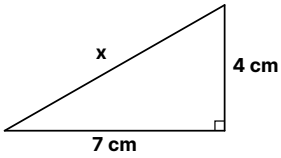
b)



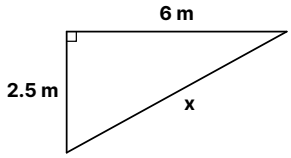
c)



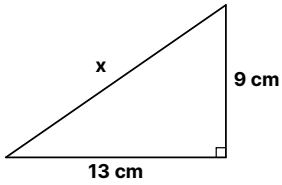
d)



e)



f)



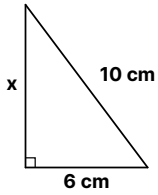
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C Finding a shorter side ●●○

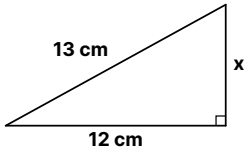
⚠ **Watch out:** Here the hypotenuse is given, so subtract: $a^2 = c^2 - b^2$.

3. Find the length of the side marked x. Round to 1 decimal place where needed.

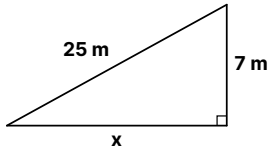
a)



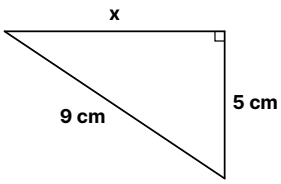
b)



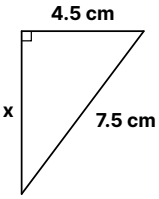
c)



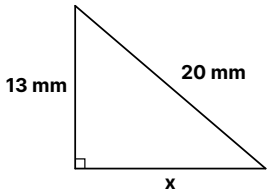
d)



e)



f)



D Is it a right-angled triangle? ●●○

Remember: It is right-angled only if the squares of the two shorter sides add up to the square of the longest side.

4. These are the side lengths of six triangles. Write yes if the triangle is right-angled and no if it is not.

a) 6 cm, 8 cm, 10 cm _____ **b)** 5 cm, 6 cm, 8 cm _____ **c)** 9 m, 40 m, 41 m _____

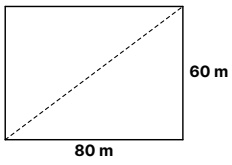
d) 7 m, 10 m, 12 m _____ **e)** 1.5 cm, 2 cm, 2.5 cm _____ **f)** 10 mm, 15 mm, 18 mm _____

E Problems ●●●

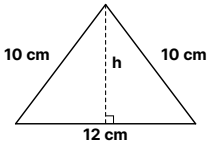
5. A 5 m ladder leans against a wall. The foot of the ladder is 1.5 m from the wall. How high up the wall does the ladder reach?



6. A rectangular field is 80 m long and 60 m wide. A path runs straight across it from one corner to the opposite corner. How long is the path?



7. An isosceles triangle has two sides of 10 cm and a base of 12 cm. Find its height, h.



★ Extension

8. Find the distance between the points $(1, 2)$ and $(7, 10)$.

9. A right-angled triangle has a hypotenuse of 25 cm and one side of 7 cm. Find its area.

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Pythagoras' Theorem

Name _____ Date _____ Calculator allowed Version 2

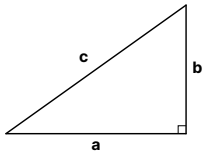
Scan to practise this topic online

KEY WORDS hypotenuse *(tapa roa)* · right-angled triangle *(tapatoru hāngai)* · square root *(pūtakerua)*

A The hypotenuse ●○○

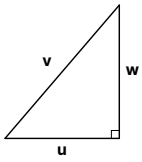
1. Write the letter of the hypotenuse of each triangle.

a)



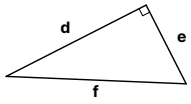
Hypotenuse: _____

b)



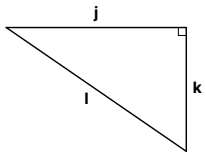
Hypotenuse: _____

c)



Hypotenuse: _____

d)



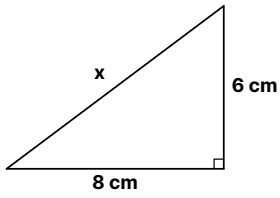
Hypotenuse: _____

B Finding the hypotenuse ●○○

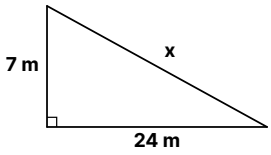
> **Remember:** $a^2 + b^2 = c^2$, where c is the hypotenuse, the longest side.

2. Find the hypotenuse, x. Round to 1 decimal place where needed. Diagrams are not to scale.

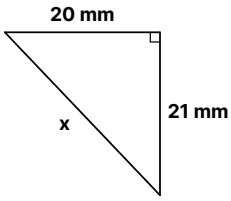
a)



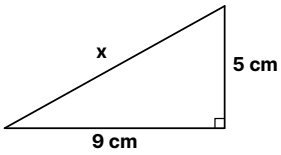
b)



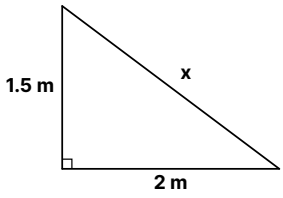
c)



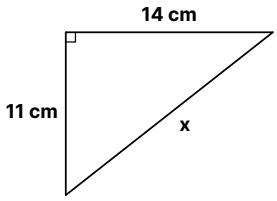
d)



e)



f)



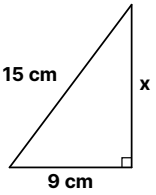
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C Finding a shorter side ●●○

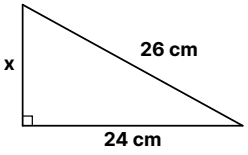
⚠ **Watch out:** Here the hypotenuse is given, so subtract: $a^2 = c^2 - b^2$.

3. Find the length of the side marked x. Round to 1 decimal place where needed.

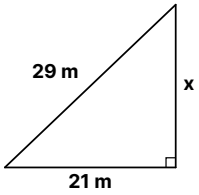
a)



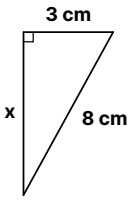
b)



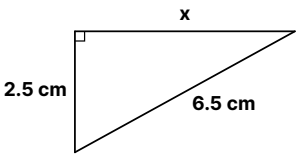
c)



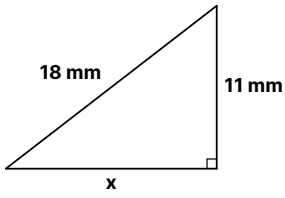
d)



e)



f)



D Is it a right-angled triangle? ●●○

Remember: It is right-angled only if the squares of the two shorter sides add up to the square of the longest side.

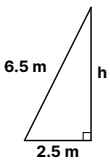
4. These are the side lengths of six triangles. Write yes if the triangle is right-angled and no if it is not.

a) 9 cm, 12 cm, 15 cm _____ **b)** 4 cm, 7 cm, 8 cm _____ **c)** 11 m, 60 m, 61 m _____

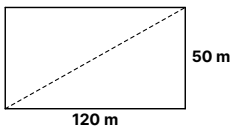
d) 8 m, 11 m, 14 m _____ **e)** 2.5 cm, 6 cm, 6.5 cm _____ **f)** 12 mm, 16 mm, 21 mm _____

E Problems ●●●

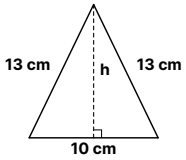
5. A 6.5 m ladder leans against a wall. The foot of the ladder is 2.5 m from the wall. How high up the wall does the ladder reach?



6. A rectangular field is 120 m long and 50 m wide. A path runs straight across it from one corner to the opposite corner. How long is the path?



7. An isosceles triangle has two sides of 13 cm and a base of 10 cm. Find its height, h.



★ Extension

8. Find the distance between the points $(2, 1)$ and $(14, 6)$.

9. A right-angled triangle has a hypotenuse of 17 cm and one side of 8 cm. Find its area.

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